Phasors

- a phasor is a complex (or analytic) representation of a wave function.
- a very useful technique for calculating resultant waves.
- phasors enable waves to be represented in a vector form.

Phasors

⇒ Recall the wave equation:

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$$

⇒ we know that one solution form is:

$$y(x,t) = y_m \sin(kx + \omega t + \phi)$$

⇒ another solution form is analytic:

$$y(x,t) = y_1 + i y_2 \quad \text{where}$$

$$y_1 = y_m \cos(kx + \omega t + \phi)$$

$$y_2 = y_m \sin(kx + \omega t + \phi)$$

Let's explore further

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Phasors

$$y = y_1 + i y_2$$

$$= y_m \left[\cos(kx + \omega t + \phi) + i \sin(kx + \omega t + \phi) \right]$$

⇒ We can use Euler's Identity to simplify this expression:

$$e^{i\theta} = \cos\theta + i \sin\theta$$

$$y(x,t) = y_m e^{i(kx + \omega t + \phi)}$$

$$= y_m e^{ikx} e^{i(\omega t + \phi)}$$

$$= \underbrace{x_m e^{i(\omega t + \phi)}}_{\tilde{x}(t)}$$

$\tilde{x}(t)$ Phasor → time dependent

Phasors

$$\tilde{x}(t) = x_m e^{i(\omega t + \phi)}$$

$\Rightarrow$ We can define displacement as the real part.

$$Re(\tilde{x}(t)) = x_m \cos(\omega t + \phi)$$

- Another way to represent the phasor:

$$\tilde{x}(t) = x_m e^{i\omega t} e^{i\phi} = \underbrace{x_m e^{i\phi}}_{\tilde{x}_m} e^{i\omega t}$$

$$\tilde{x}(t) = \tilde{x}_m e^{i\omega t} \quad \text{where}$$

$$\tilde{x}_m = x_m e^{i\phi}$$

Phasors

- Let's use phasors to determine the resultant wave from wave superposition.

wave 1: $x_1 = x_{m_1} \cos(\omega t + \phi_1)$

wave 2: $x_2 = x_{m_2} \cos(\omega t + \phi_2)$

$\Rightarrow$ Convert to complex form:

$\tilde{x}_1 = \tilde{x}_{m_1} e^{i\omega t}$ where $\tilde{x}_{m_1} = x_{m_1} e^{i\phi_1}$

$\tilde{x}_2 = \tilde{x}_{m_2} e^{i\omega t}$ where $\tilde{x}_{m_2} = x_{m_2} e^{i\phi_2}$

$\Rightarrow$ Superposition by summation:

$$\tilde{x}_R = \tilde{x}_1 + \tilde{x}_2$$

$$= \underbrace{(\tilde{x}_{m_1} + \tilde{x}_{m_2})}_{\tilde{x}_m} e^{i\omega t}$$

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Phasors.

$$\tilde{x}_R = \tilde{x}_m e^{i\omega t} \quad \text{where}$$

$$\tilde{x}_m = |\tilde{x}_m| e^{i\phi}$$

$|\tilde{x}_m|$: amplitude of the superposition of the 2 phasors.

ϕ: phase constant of the superposition

Practice Problem 1

Consider 2 waves travelling along the same string.

$$y_1(x,t) = 5.6\text{m} \sin(2\pi x - 400\pi t)$$

$$y_2(x,t) = 6.6\text{m} \sin(2\pi x - 400\pi t + 0.8\pi \text{ rad})$$

Determine the following properties of the resultant wave:

i) the wave amplitude

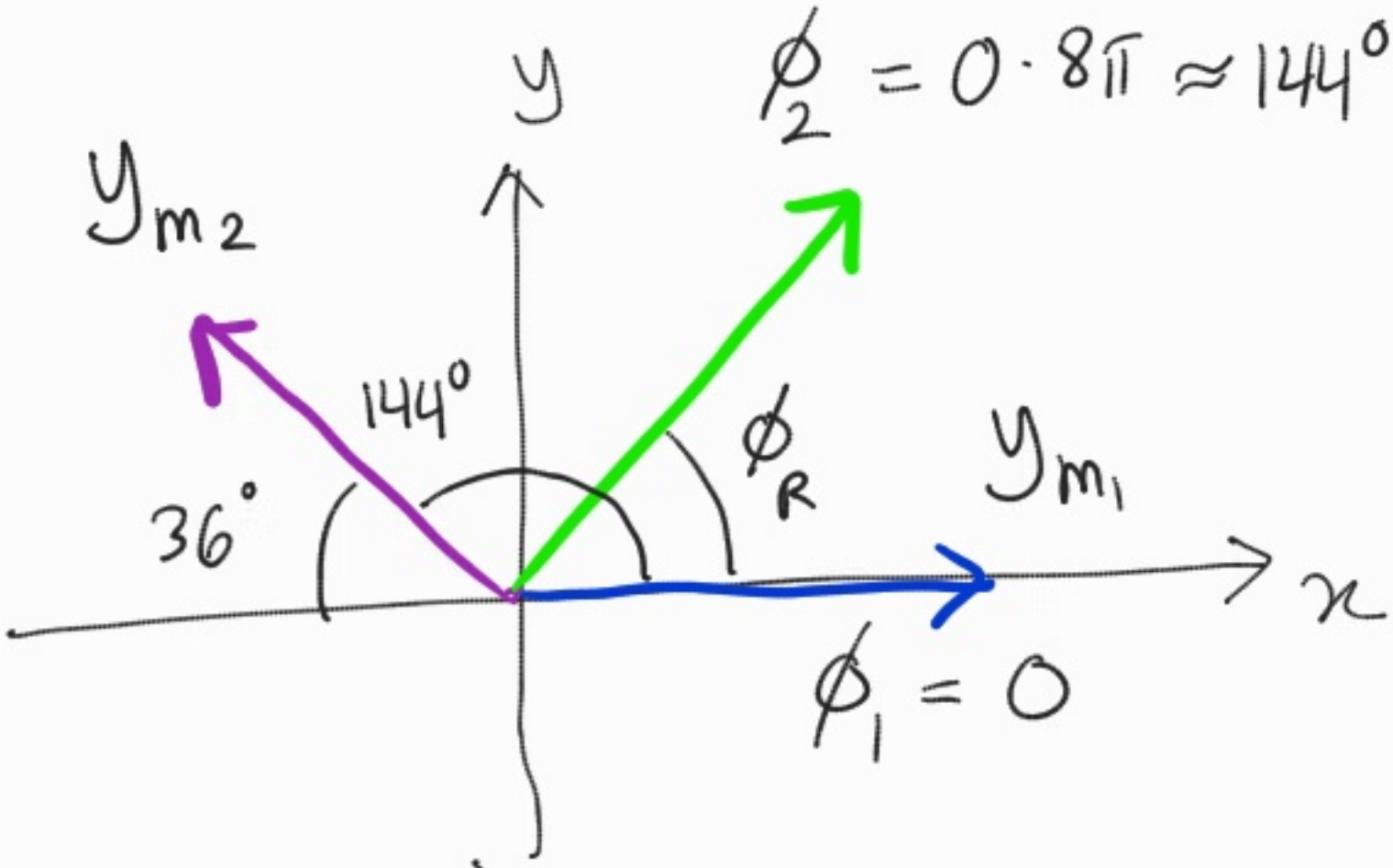
ii) the phase angle with respect to wave 1.

Solution :

$$y_1(x,t) = 5.6\text{m} \sin(2\pi x - 400\pi t)$$

$$y_2(x,t) = 6.6\text{m} \sin(2\pi x - 400\pi t + 0.8\pi \text{ rad})$$

$$\phi_2 = 0.8\pi \approx 144^\circ$$



For the x-axis's :

$$y_{m_x} = (y_{m_1})_x + (y_{m_2})_x$$

$$= [5.6 - 6.6 \cos 36^\circ]\text{m}$$

$$= 0.261\text{ m}$$

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Solution (Con't)

For the y-axis:

$$y_{m_y} = \cancel{(y_{m_1})_y}^{\,0} + (y_{m_2})_y$$

$$= [6.6 \sin 36^\circ]\,m$$

$$= 3.88\,m$$

$$y_m = \sqrt{y_{m_x}^2 + y_{m_y}^2}$$

$$= \sqrt{0.261^2 + 3.88^2}$$

$$= \underline{3.89\,m}$$

$$\phi_R = \tan^{-1}\left(\frac{3.88m}{0.261m}\right) \approx \underline{86.15^\circ}$$

Practice Problem 1 [Con't]

- What if a 3rd wave is introduced on the string in the same direction of the first 2 waves. This 3rd wave has an amplitude of 5 m. What should be its phase angle, ϕ_3 in order to maximize the amplitude of the resulting wave?

Solution :

- To maximize the amplitude of the resultant wave the 3rd wave should be in phase with the resultant wave of the first 2 waves.

$$\phi_3 = \phi_R$$

Practice Problem 2

Determine the phasor representation of the following electric current:

$$i(t) = -4\cos\left(\omega t + \frac{3\pi}{4}\right)$$

Solution:

$$i(t) = -4\cos\left(\omega t + \frac{3\pi}{4}\right)$$

$$\tilde{i}(t) = -4\, e^{j\left(\omega t + \frac{3\pi}{4}\right)}$$

Since $e^{-j\pi} = -1$, we can simplify further

$$\tilde{i}(t) = 4\, e^{-j\pi}\, e^{j\left(\omega t + \frac{3\pi}{4}\right)}$$

$$= 4\, e^{j\left(\omega t - \frac{\pi}{4}\right)}$$

$$= 4\, e^{-j\pi/4}\, e^{j\omega t}$$

$$= \underbrace{4\, e^{-j\pi/4}}\; e^{j\omega t}$$

$$\tilde{I}_m$$

$$\tilde{i}(t) = \tilde{I}_m\, e^{j\omega t} \quad \text{where}$$

$$\tilde{I}_m = 4e^{-j\pi/4}$$

Practice Problem 3

Determine the superposition of the following electric currents using phasors:

$$i_1(t) = 5 \sin(\omega t + \pi/4)$$

$$i_2(t) = 7 \cos(\omega t - \pi/4)$$

Solution.

$$i_1(t) = 5 \sin(\omega t + \pi/4)$$

$$i_2(t) = 7 \cos(\omega t - \pi/4)$$

⇒ Let's convert $i_1(t)$ to a cosine form via a phase shift.

$$\cos\left(\theta - \frac{\pi}{2}\right) = \sin\theta$$

$$i_1(t) = 5 \cos(\omega t - \pi/4)$$

$$\tilde{i}_1(t) = 5 e^{-j\pi/4} e^{j\omega t}$$

$$\tilde{i}_2(t) = 7 e^{-j\pi/4} e^{j\omega t}$$

$$\tilde{i}(t) = \tilde{i}_1(t) + \tilde{i}_2(t)$$

$$\tilde{i}(t) = \underbrace{12 e^{-j\pi/4}}_{\tilde{i}_m} e^{j\omega t}$$